



Final Assignment

Luís Paulo Santos, Dezembro, 2025

Develop and assess a full pipeline from simulation to visualization.

1. Identify a problem to simulate which:
 - a. generates data appropriate for scientific visualization;
 - b. has code publicly available that you can tamper with;
2. Identify which insights you want to get about the problem and its data, which will guide your data exploration through visualization;
3. Instrument the code, such that you can output the data to Paraview;
4. Develop your visualization pipeline;
5. Qualitatively assess and discuss how the proposed visualization allows you to get insight about the problem being analysed;
6. Qualitatively assess both the simulation and visualization performance as the data volume grows;
7. Document your project and results.

Optional:

8. Demonstrate if and how your simulation code can benefit from parallel processing at some level (multi-threading, multiple nodes,...)
9. Demonstrate if and how your visualization process can benefit from parallel visualization by using multiple pvpython processes

Deliverables

Prepare a git repository that should include:

- the simulation code and building configuration files (make, cmake, ...)
- the visualization pipeline (using the Paraview's "Save State" option)
- a data set generated with the simulator ready to be used in Paraview
- an image (or animation) of the visualization
- a report (max 10 pages) describing the goals, the process and the results.

The repository's address, together with the group members' identifications (max 3 members), must be submitted in the form made available in the e-learning platform until the 1st of February, 2026, 23h59m.